

CYCLIC NORM CERTIFICATES FOR EQUATIONAL JACOBSON PROOFS IN THE p^3 -CASE

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ABSTRACT. Jacobson’s commutativity theorem implies that a ring in which each element is potent is commutative, but the theorem does not by itself produce explicit equational derivations for fixed exponents. We study the first prime-power layer not covered by Brandenburg’s explicit $k = 1$ and $k = 2$ arguments: characteristic- p rings satisfying $x^{p^3} = x$. Brandenburg’s period criterion proves the case $\gcd(3, p^3 - 1) = 1$, leaving exactly the primes $p \equiv 1 \pmod{3}$. For this remaining family we prove finite certificates for the two monomial constructive Wedderburn statements $W_{p,3,Tp}$ and $W_{p,3,Tp^2}$. The new step is a cyclic norm argument. On an irreducible cubic Frobenius component, a unit b satisfying $ba = a^{p^i}b$ has b^3 fixed; a finite norm section supplies u with $u\sigma(u)\sigma^2(u) = b^3$. The normalized element $u^{-1}b$ has cube 1, and when $p \equiv 1 \pmod{3}$, split-polynomial centrality in reduced rings forces it to be central, collapsing the Frobenius cycle. Exact symbolic checks verify the congruence split and the finite norm-component structure for representative primes. Thus, for every fixed prime p , p^3 -rings have equational commutativity certificates.

1. INTRODUCTION

Jacobson’s commutativity theorem says that a ring is commutative if every element x satisfies $x^n = x$ for some integer $n > 1$ depending on x [6]. The theorem is classical, but the fixed-exponent form raises a more syntactic question. If n is fixed and every element satisfies $x^n = x$, can one derive $xy = yx$ by equations alone, starting from the ring axioms and the identity $x^n = x$?

This question is not only a reformulation of a known theorem. For fixed n , Birkhoff’s completeness theorem guarantees that such a derivation exists once the semantic theorem is known, but it gives no useful derivation [1]. Explicit proofs are delicate: they must replace quotient arguments, ideals, and ordinary case distinctions by finite polynomial identities. Work of Herstein, Luh, and Dolan gives elementary structural proofs of Jacobson-type commutativity theorems [5, 8, 4]. Recent fixed-exponent proofs and Prover9-assisted centrality arguments appear in Kinyon–MacHale [7], while computational approaches to fixed identities appear in Burris–Lawrence and Wavrik [3, 9].

Brandenburg recently gave a systematic equational program for this problem [2]. The fixed-exponent problem is reduced to prime-power identities in prime characteristic. In characteristic p , one studies p^k -rings, meaning rings satisfying $px = 0$ and $x^{p^k} = x$. Brandenburg proves the cases $k = 1$ and $k = 2$, reduces the general case to constructive Wedderburn statements, and proves a period criterion

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that handles many higher prime powers. For $k = 3$, that criterion proves all primes with

$$\gcd(3, p^3 - 1) = 1.$$

Equivalently, it covers $p = 2$, $p = 3$, and primes $p \equiv 2 \pmod{3}$. The first family not covered by this criterion is $p \equiv 1 \pmod{3}$.

We prove that this remaining family is also controlled by finite equational certificates. By Brandenburg's monomial reduction, it is enough to prove the two statements

$$W_{p,3,T^p} \quad \text{and} \quad W_{p,3,T^{p^2}},$$

where $W_{p,3,f}$ asserts that $ba = f(a)b$ implies $ba = ab$ in every p^3 -ring. The obstruction in the missing congruence class is a genuine Frobenius 3-cycle on irreducible cubic components. The main observation of this paper is that the cycle can be absorbed by a cyclic norm section.

The proof has three ingredients. First, if a reduced ring element satisfies a separable polynomial split over $\mathbb{F}_p$, then it is central. In particular, when $p \equiv 1 \pmod{3}$, every element satisfying $c^3 = 1$ is central. Second, on each irreducible cubic component, a finite quotient algebra supplies a polynomial $U(S, T)$ whose cyclic norm is S . Third, if $ba = a^{p^i}b$ and b is a unit on its central support, then $s = b^3$ is fixed by the Frobenius cycle. Choosing $u = U(s, a)$, the element $c = u^{-1}b$ satisfies $c^3 = 1$, hence is central; this forces $a = a^{p^i}$, contradicting the nontrivial cubic component unless the component vanishes.

Our contributions are:

- We isolate a split-polynomial centrality lemma that turns $c^3 = 1$ into centrality exactly in the congruence class $p \equiv 1 \pmod{3}$.
- We prove a finite cyclic norm section for every irreducible cubic factor of $T^{p^3} - T$ and for both Frobenius generators $T \mapsto T^p$ and $T \mapsto T^{p^2}$.
- We combine these ingredients with Brandenburg's reductions to obtain finite equational certificates for all characteristic- p rings satisfying $x^{p^3} = x$.
- We record exact symbolic checks that reproduce the congruence split and verify the finite component structure used by the norm argument.

Section 2 fixes notation and recalls the reductions from Brandenburg that we use. Section 3 proves the centrality, norm-section, and monomial Wedderburn results, and then combines them into the $k = 3$ theorem. Section 4 discusses the certificate interpretation and open computational questions. Section 5 summarizes the result.

2. PRELIMINARIES

All rings in this paper are associative and unital. If p is prime, $\mathbb{F}_p$ denotes the field with p elements. We write $Z(A)$ for the center of a ring A . For the main problem we put

$$q = p^3.$$

Definition 2.1. A p^3 -ring is a ring A such that

$$px = 0 \quad \text{and} \quad x^{p^3} = x$$

for every $x \in A$.

Lemma 2.2. *Every p^3 -ring is reduced, and every idempotent in a reduced ring is central.*

Proof. If A is a p^3 -ring and $x^2 = 0$, then $x^{p^3} = 0$, so $x = 0$. Thus A has no nonzero square-zero elements. If y is nilpotent, then $y^{2^r} = 0$ for some r , and descending from $y^{2^r} = 0$ shows $y = 0$. Hence A is reduced.

Let A be reduced and let $e^2 = e$. For any $x \in A$,

$$(ex - exe)^2 = 0 \quad \text{and} \quad (xe - exe)^2 = 0.$$

Reducedness gives $ex = exe = xe$. Hence $e \in Z(A)$. $\square$

If $x^q = x$, then positive powers of x depend only on the exponent modulo $q - 1$: for $r \geq 1$,

$$x^{r+q-1} = x^r.$$

In particular, if x is a unit in a p^3 -ring, then $x^{q-1} = 1$.

Definition 2.3. Let $f \in \mathbb{F}_p[T]$. The constructive Wedderburn statement $W_{p,3,f}$ is the assertion that, in every p^3 -ring A , for all $a, b \in A$,

$$ba = f(a)b \implies ba = ab.$$

We use two reductions from Brandenburg [2]. First, for $k = 3$, commutativity of all p^3 -rings follows from equational proofs of the two monomial statements

$$W_{p,3,T^p} \quad \text{and} \quad W_{p,3,T^{p^2}}.$$

Second, Brandenburg's period criterion proves these monomial statements whenever $\gcd(3, p^3 - 1) = 1$. Since

$$\gcd(3, p^3 - 1) = \begin{cases} 1, & p = 2, p = 3, \text{ or } p \equiv 2 \pmod{3}, \\ 3, & p \equiv 1 \pmod{3}, \end{cases}$$

the only case not settled by the period criterion is $p \equiv 1 \pmod{3}$.

We also use the standard factorization

$$T^q - T = \prod_{\alpha \in \mathbb{F}_q} (T - \alpha)$$

over $\mathbb{F}_q$. Over $\mathbb{F}_p$, this is the product of all monic irreducible polynomials whose degrees divide 3. Thus the irreducible factors are linear factors and irreducible cubic factors. For each irreducible factor g of $T^q - T$, Bezout identities for the pair g and $(T^q - T)/g$ give a polynomial idempotent $e_g(T)$ modulo $T^q - T$. If $a \in A$, then $e_g(a)$ is a central idempotent by Lemma 2.2, and the idempotents $e_g(a)$ decompose A into the usual linear and cubic Frobenius components.

Definition 2.4. A *finite equational certificate* for a statement about p^3 -rings is a finite list of polynomial identities over $\mathbb{F}_p$, together with the indicated substitutions into the ring identities, whose validity implies the desired equation by equational logic. In this paper the certificates are finite because, for fixed p , there are only finitely many irreducible cubic factors of $T^{p^3} - T$, and each norm identity is verified in a finite-dimensional quotient of $\mathbb{F}_p[S, T]$.

Remark 2.5. The proofs below are written in ordinary ring-theoretic language for readability. Each use of a component, a support, or a polynomial inverse is represented by an explicit idempotent or Bezout identity. Thus the semantic notation records a finite equational derivation rather than adding an extra model-theoretic assumption.

3. MAIN RESULTS

We begin with a centrality lemma for elements annihilated by split squarefree polynomials.

Lemma 3.1 (Split-polynomial centrality). *Let A be a reduced $\mathbb{F}_p$ -algebra. Let*

$$m(T) = \prod_{\lambda \in \Lambda} (T - \lambda)$$

with $\Lambda \subseteq \mathbb{F}_p$ finite and with distinct elements. If $m(c) = 0$ for some $c \in A$, then $c \in Z(A)$. In particular, if $p \equiv 1 \pmod{3}$ and $c^3 = 1$, then $c \in Z(A)$.

Proof. For each $\lambda \in \Lambda$, let $e_\lambda(T) \in \mathbb{F}_p[T]$ be the Lagrange interpolation polynomial satisfying

$$e_\lambda(\mu) = \begin{cases} 1, & \mu = \lambda, \\ 0, & \mu \neq \lambda \end{cases} \quad (\mu \in \Lambda).$$

Modulo $m(T)$, the polynomials e_λ are pairwise orthogonal idempotents and

$$1 = \sum_{\lambda \in \Lambda} e_\lambda(T), \quad T = \sum_{\lambda \in \Lambda} \lambda e_\lambda(T).$$

Evaluating at c , we obtain pairwise orthogonal idempotents $e_\lambda(c)$. By Lemma 2.2, each $e_\lambda(c)$ is central. Therefore

$$c = \sum_{\lambda \in \Lambda} \lambda e_\lambda(c)$$

is central.

If $p \equiv 1 \pmod{3}$, then $p \neq 3$ and $\mathbb{F}_p^\times$ contains the three cube roots of unity. Hence $T^3 - 1$ splits into distinct linear factors over $\mathbb{F}_p$. Applying the first part to $m(T) = T^3 - 1$ proves the final assertion. $\square$

The next lemma is the finite algebra input used to normalize a Frobenius-skew unit.

Lemma 3.2 (Finite cyclic norm section). *Let $p \equiv 1 \pmod{3}$, let $q = p^3$, let $g \in \mathbb{F}_p[T]$ be irreducible of degree 3, and let $i \in \{1, 2\}$. Put*

$$B = \mathbb{F}_p[S, T] / (S^{q-1} - 1, g(T)).$$

Let σ_i be the $\mathbb{F}_p$ -algebra automorphism of B defined by

$$\sigma_i(S) = S, \quad \sigma_i(T) = T^{p^i}.$$

Then there is a unit $U \in B$ such that

$$U \sigma_i(U) \sigma_i^2(U) = S.$$

Proof. Let

$$C = \mathbb{F}_p[S] / (S^{q-1} - 1).$$

The polynomial $S^{q-1} - 1$ is squarefree in characteristic p , and its roots are exactly $\mathbb{F}_q^\times$. Therefore every simple factor of C is a finite field whose degree over $\mathbb{F}_p$ divides 3. Since 3 is prime, the degrees are 1 or 3.

We prove the asserted norm-surjectivity on each simple factor of C . If S maps to a degree-one factor, then $S = s_0 \in \mathbb{F}_p^\times$, and

$$B \otimes_C \mathbb{F}_p \cong \mathbb{F}_p[T] / (g(T)) \cong \mathbb{F}_q.$$

Under this identification, σ_i is a generator of $\text{Gal}(\mathbb{F}_q/\mathbb{F}_p)$. The norm

$$x \mapsto x \sigma_i(x) \sigma_i^2(x)$$

maps $\mathbb{F}_q^\times$ onto $\mathbb{F}_p^\times$, so s_0 has a preimage.

If S maps to a degree-three factor, then that factor is a copy $K \cong \mathbb{F}_q$. The polynomial g splits over K , so

$$B \otimes_C K \cong K^3.$$

The automorphism σ_i fixes K and cyclically permutes the three factors, because $T \mapsto T^{p^i}$ is a 3-cycle on the roots of the irreducible cubic g . Hence the cyclic norm sends

$$(x_0, x_1, x_2) \mapsto (x_0 x_1 x_2, x_0 x_1 x_2, x_0 x_1 x_2).$$

This map is surjective onto the diagonal copy of $K^\times$; in particular it has a preimage of the element represented by S .

The Chinese remainder theorem combines these choices over all simple factors of C . The resulting element $U \in B$ has nonzero image in every simple component, hence is a unit, and satisfies

$$U \sigma_i(U) \sigma_i^2(U) = S$$

in B . □

Proposition 3.3. *Let $p \equiv 1 \pmod{3}$. Then the constructive Wedderburn statements $W_{p,3,T^p}$ and $W_{p,3,T^{p^2}}$ admit finite equational certificates.*

Proof. It is enough to treat $W_{p,3,T^{p^i}}$ for a fixed $i \in \{1, 2\}$. Let A be a p^3 -ring, put $q = p^3$, and suppose

$$ba = a^{p^i} b$$

for $a, b \in A$. Write $\sigma(a) = a^{p^i}$.

First reduce to the central support on which b is a unit. Let $e = b^{q-1}$. Then e is a central idempotent, $eb = b$, and $(1-e)b = 0$. On the component eA , the element b is a unit with inverse b^{q-2} . On the complementary component, $b = 0$, so $ba = ab = 0$. Thus we may replace A by eA and assume that b is a unit.

Decompose a using the central idempotents attached to the irreducible factors of $T^q - T$. On a linear component, $a \in \mathbb{F}_p$ in the polynomial sense, so $a^{p^i} = a$, and the desired commutativity follows immediately from $ba = a^{p^i} b$.

It remains to consider a component on which

$$g(a) = 0$$

for an irreducible cubic $g \in \mathbb{F}_p[T]$. In this component, $a^{p^i} - a$ is a unit. Indeed, g is coprime to $T^{p^i} - T$: a common root would lie in $\mathbb{F}_{p^i}$, while a root of g has degree 3 over $\mathbb{F}_p$. A Bezout identity therefore gives a polynomial inverse for $a^{p^i} - a$.

We now show that such a cubic component must be zero. Iterating the relation $ba = \sigma(a)b$ gives

$$b^3 a = \sigma^3(a) b^3 = ab^3,$$

since $\sigma^3(a) = a^{p^{3i}} = a$ in a p^3 -ring. Let $s = b^3$. Then s commutes with a , and it also commutes with b . Since s is a unit, $s^{q-1} = 1$.

Apply Lemma 3.2 to the cubic g . Under the homomorphism

$$\mathbb{F}_p[S, T]/(S^{q-1} - 1, g(T)) \longrightarrow A, \quad S \mapsto s, \quad T \mapsto a,$$

the norm-section element U maps to an element u in the commutative subring generated by s and a , and

$$u \sigma(u) \sigma^2(u) = s.$$

Here σ acts on this subring by fixing s and sending a to a^{p^i} . The displayed identity makes u a unit. Since $bu = \sigma(u)b$, the element

$$c = u^{-1}b$$

satisfies

$$c^3 = u^{-1} \sigma(u)^{-1} \sigma^2(u)^{-1} b^3 = (u \sigma(u) \sigma^2(u))^{-1} s = 1.$$

By Lemma 3.1, c is central, because $p \equiv 1 \pmod{3}$. But

$$ca = u^{-1}ba = u^{-1}a^{p^i}b = a^{p^i}c.$$

Since c is a central unit, this gives $a = a^{p^i}$. This contradicts the fact that $a^{p^i} - a$ is a unit unless the identity element of the present cubic component is zero. Hence the b -unit support has no nonzero cubic component.

We have shown that, on the support of b , all nonzero components are linear, and on those components $a^{p^i} = a$. Therefore $ba = ab$ in A .

Finally, for fixed p , every ingredient used above is finite and polynomial: the support b^{q-1} , the idempotents for the factorization of $T^q - T$, the Bezout inverse of $T^{p^i} - T$ modulo each cubic g , and the norm-section identity from Lemma 3.2. Enumerating the finitely many cubic g 's gives the desired finite equational certificate. $\square$

Theorem 3.4. *Let p be prime. Every p^3 -ring is commutative. Moreover, for each fixed p , the identity $xy = yx$ has a finite equational certificate from the identities $px = 0$ and $x^{p^3} = x$.*

Proof. If $p = 2$, $p = 3$, or $p \equiv 2 \pmod{3}$, then $\gcd(3, p^3 - 1) = 1$, so Brandenburg's period criterion proves the needed monomial constructive Wedderburn statements equationally [2]. If $p \equiv 1 \pmod{3}$, Proposition 3.3 gives finite equational certificates for $W_{p,3,T^p}$ and $W_{p,3,T^{p^2}}$. Brandenburg's monomial sufficiency theorem then gives commutativity of all p^3 -rings, again by finite equational certificates for each fixed p . $\square$

Example 3.5 (A period-criterion prime). For $p = 5$, one has $p^3 - 1 = 124$ and $\gcd(3, 124) = 1$. Thus the 5^3 -case is already handled by Brandenburg's period criterion. No cyclic norm section is needed in this congruence class.

Example 3.6 (The first norm-section prime). For $p = 7$, one has $q = 343$ and $p \equiv 1 \pmod{3}$. The polynomial $T^{343} - T$ has 7 linear factors and $(343 - 7)/3 = 112$ irreducible cubic factors over $\mathbb{F}_7$. Proposition 3.3 supplies a norm-section certificate for each of those cubic factors and for both maps $T \mapsto T^7$ and $T \mapsto T^{49}$.

Remark 3.7. Lemma 3.1 uses the splitting of the annihilating polynomial over the ground field. The proof does not say that an element on an irreducible cubic component is central. The norm argument is precisely the step that converts the Frobenius-skew unit into an element satisfying the split equation $c^3 = 1$.

TABLE 1. Exact symbolic checks used as reproducibility evidence.
The skew-model row supports the proof search but is not used as
a premise in Theorem 3.4.

Check	Output	Role
Period samples	$p = 2, 3, 5, 11, 17$ have $\gcd(3, p^3 - 1) = 1$	Confirms covered classes
Bad-class samples	$p = 7, 13, 19$ have $\gcd(3, p^3 - 1) = 3$	Confirms target classes
Norm components, $p = 7$	112 cubic S -components and 6 linear ones	Matches Lemma 3.2
Norm components, $p = 13$	728 cubic S -components and 12 linear ones	Matches Lemma 3.2
Skew model, $p = 7$	$\gcd$ degree drops from 13 to 0 after adding $t = 3$	Checks first open model

3.1. **Verification.** Checks are exact. Table 1 summarizes the outputs, which are stored in `results/k3_checks.json`.

The validation commands compile the checking scripts and compare two independent JSON runs:

```
python -m py_compile src/k3_wedderburn_checks.py \
code/finite_field_reductions.py
diff <(python src/k3_wedderburn_checks.py --json) \
<(python src/k3_wedderburn_checks.py --json)
```

The computations are deterministic; they verify the finite algebra reductions used to generate certificates, not a statistical claim.

4. DISCUSSION

The $k = 3$ case separates cleanly into two mechanisms. When $\gcd(3, p^3 - 1) = 1$, Brandenburg's period criterion collapses the relevant Frobenius orbit. When $p \equiv 1 \pmod{3}$, the orbit has genuine length 3, so a different argument is needed. The cyclic norm section supplies exactly that argument: it changes a Frobenius-skew unit b into a unit $c = u^{-1}b$ whose cube is 1. Because $T^3 - 1$ splits over $\mathbb{F}_p$ in this congruence class, reducedness forces c to be central.

The proof also clarifies what remains computational rather than conceptual. For a fixed prime $p \equiv 1 \pmod{3}$, Lemma 3.2 guarantees a polynomial representative

$$U_{g,i}(S, T) \in \mathbb{F}_p[S, T]$$

for each irreducible cubic g and each $i \in \{1, 2\}$. The identity

$$U_{g,i}(S, T)U_{g,i}(S, T^{p^i})U_{g,i}(S, T^{p^{2i}}) = S$$

is then checked in the finite quotient

$$\mathbb{F}_p[S, T]/(S^{p^3-1} - 1, g(T)).$$

Printing these polynomials for every g is not necessary for the human proof, but it is the next step toward machine-checkable certificate files in the style of Brandenburg's explicit computations.

This distinction matters. The present paper proves existence of finite certificates and identifies the finite quotient rings in which they live. It does not claim that

the expanded certificates are short. Already for $p = 7$, there are 112 irreducible cubic factors of $T^{343} - T$, so a literal certificate listing can be large. Symmetries among cubic factors, reciprocal polynomials, and linear substitutions should reduce the output, but we have not optimized it here.

Several questions remain open.

- (1) Build a certificate generator that computes $U_{g,i}(S, T)$ for every relevant g and emits a machine-checkable equational derivation.
- (2) Minimize the resulting certificates using the action of Frobenius, reciprocal transformations, and linear changes of variable.
- (3) Determine whether the cyclic norm method extends to higher layers p^k , where the bad congruence classes contain longer Frobenius cycles.
- (4) Compare the norm certificates with automated equational searches of the kind developed for finite-field rewrite systems and noncommutative polynomial identities [3, 9].

5. CONCLUSION

We proved that the p^3 -case of the fixed-exponent Jacobson problem admits finite equational certificates for every prime p . Brandenburg's period criterion covers $p = 2$, $p = 3$, and $p \equiv 2 \pmod{3}$. For the remaining primes $p \equiv 1 \pmod{3}$, the cyclic norm argument proves the two monomial constructive Wedderburn statements required by Brandenburg's reduction.

The essential new step is the normalization of a Frobenius-skew unit b . On each irreducible cubic component, the element b^3 is fixed by the cycle; a finite norm section writes it as $u\sigma(u)\sigma^2(u)$, and then $u^{-1}b$ has cube 1. Since $T^3 - 1$ splits in the remaining congruence class, reducedness makes this normalized element central and eliminates the cubic component. The proof gives a concrete route from the semantic commutativity theorem to explicit polynomial certificates for each fixed exponent p^3 .

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